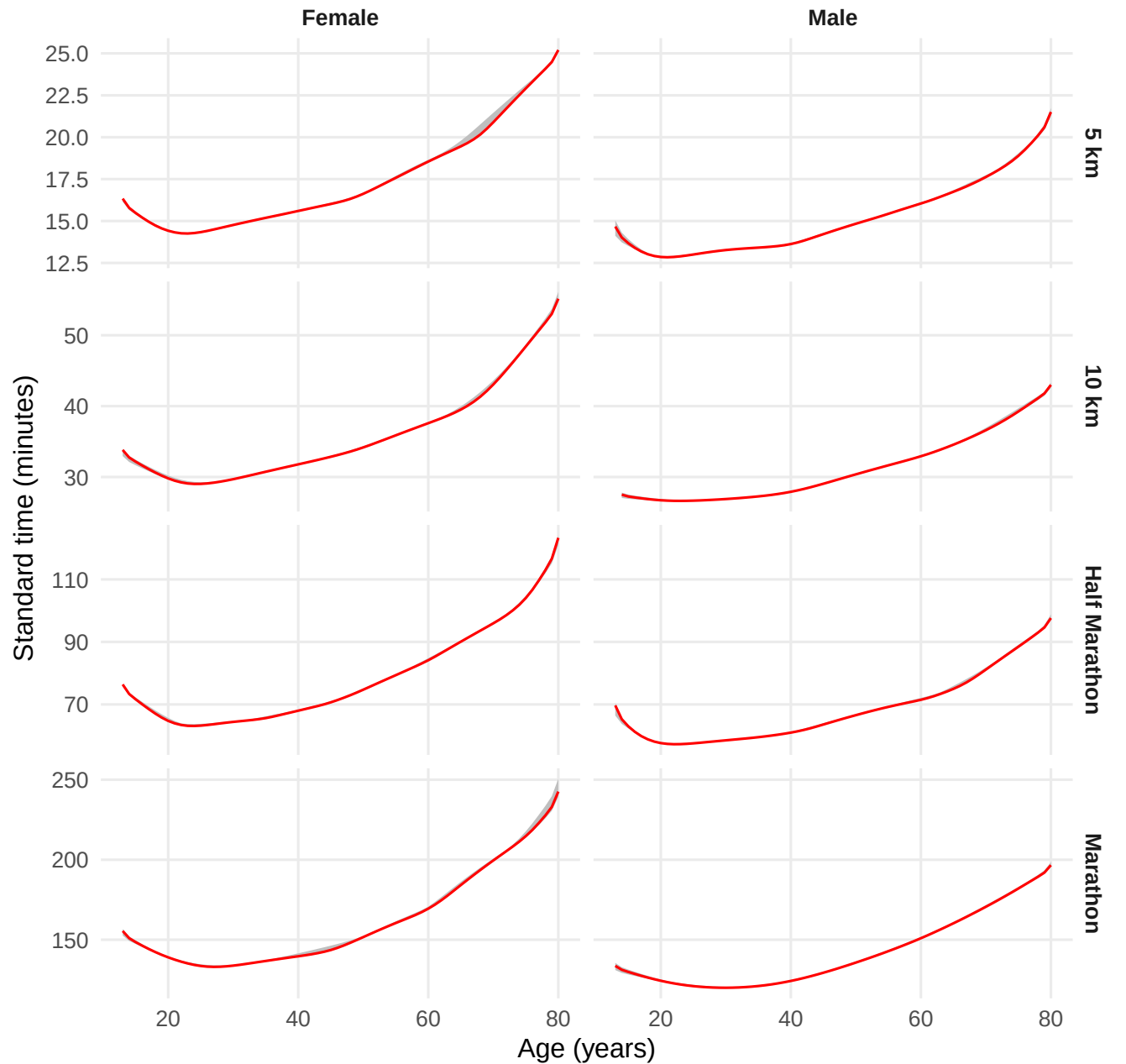
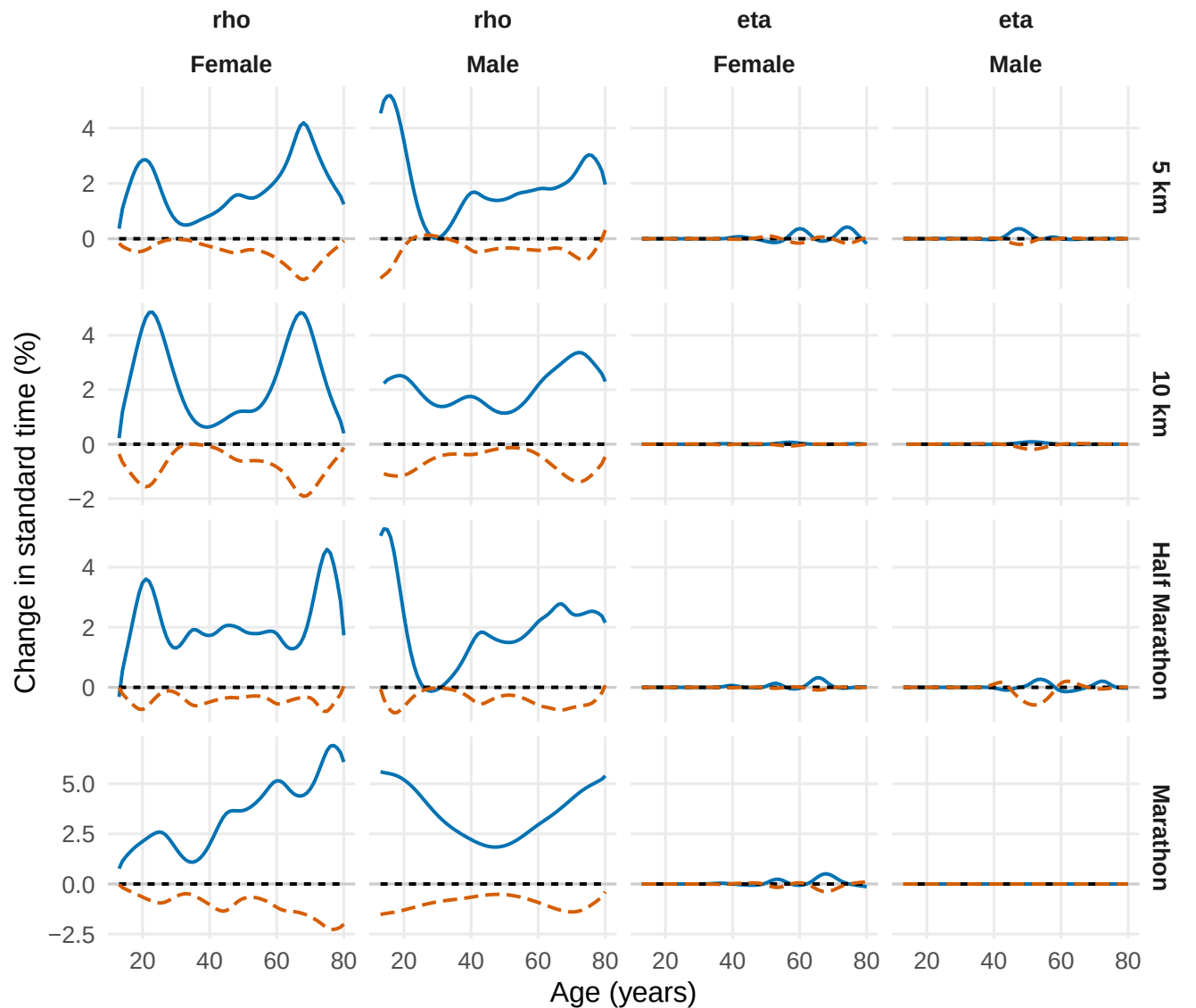


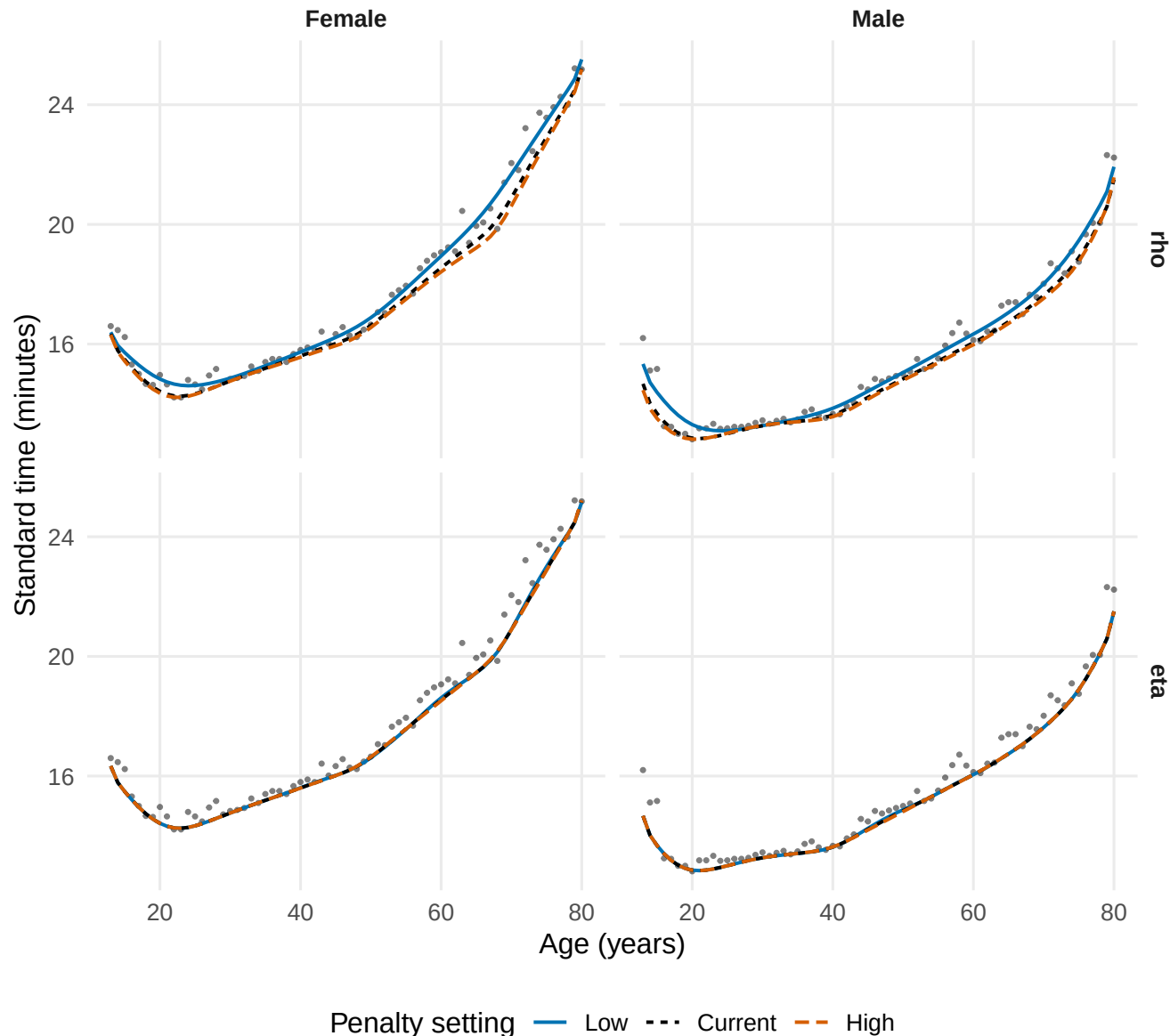


Delete-one envelope at fixed lambda; not a 95% CI



rho: 1, 30, 1000 (eta = 1000); eta: 0, 1000, 1000000 (rho = 30).  
 Lambda fixed at the current fit's CV choice in each sex–event group.

# 5 km: sensitivity to penalty settings



rho: 1, 30, 1000 (eta = 1000); eta: 0, 1000, 1000000 (rho = 30).  
 Lambda fixed at the current fit's CV choice in each sex-event group.

Female

Male

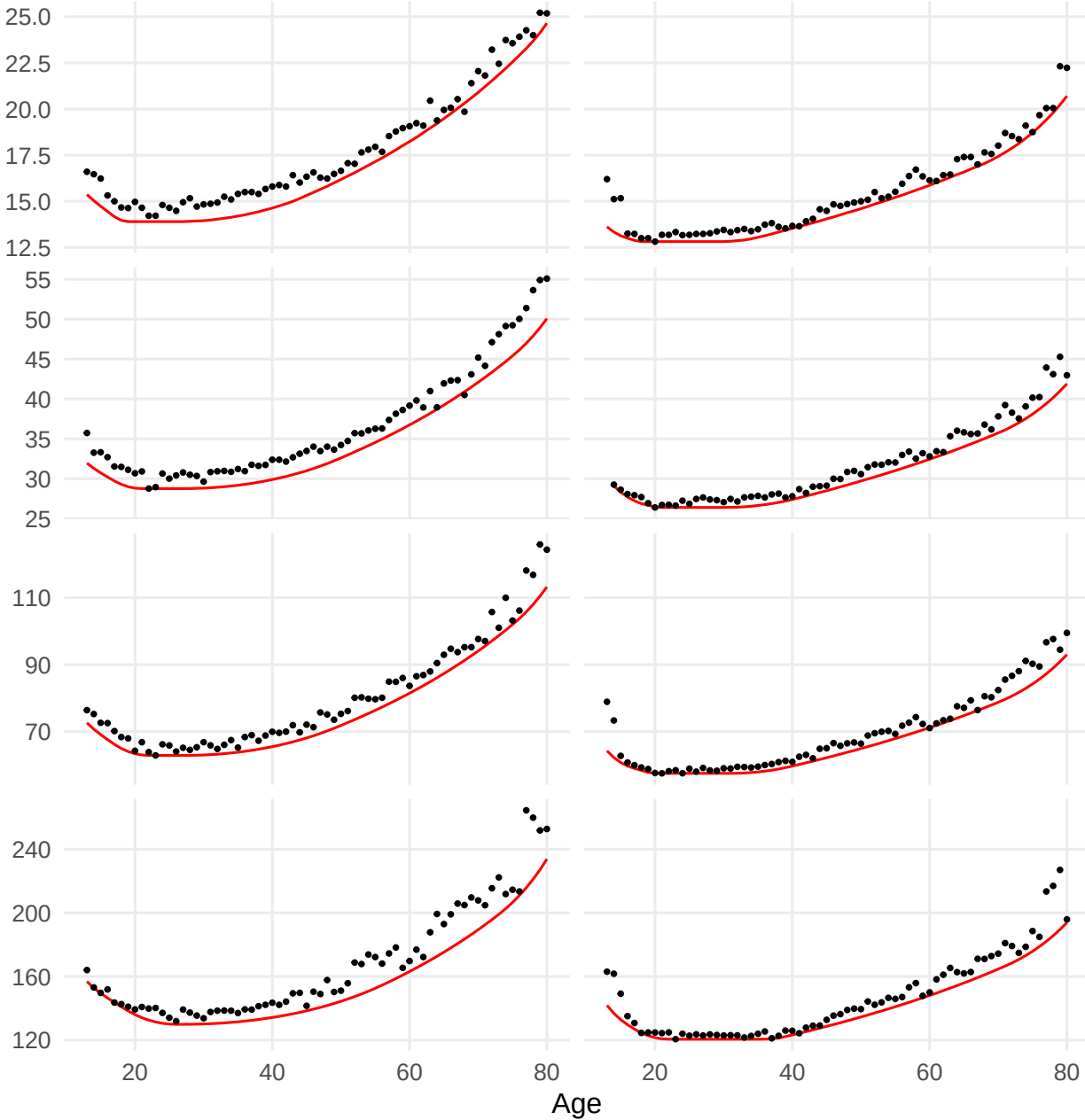
5 km

10 km

Half Marathon

Marathon

Time (minutes)



Female

Male

5 km

10 km

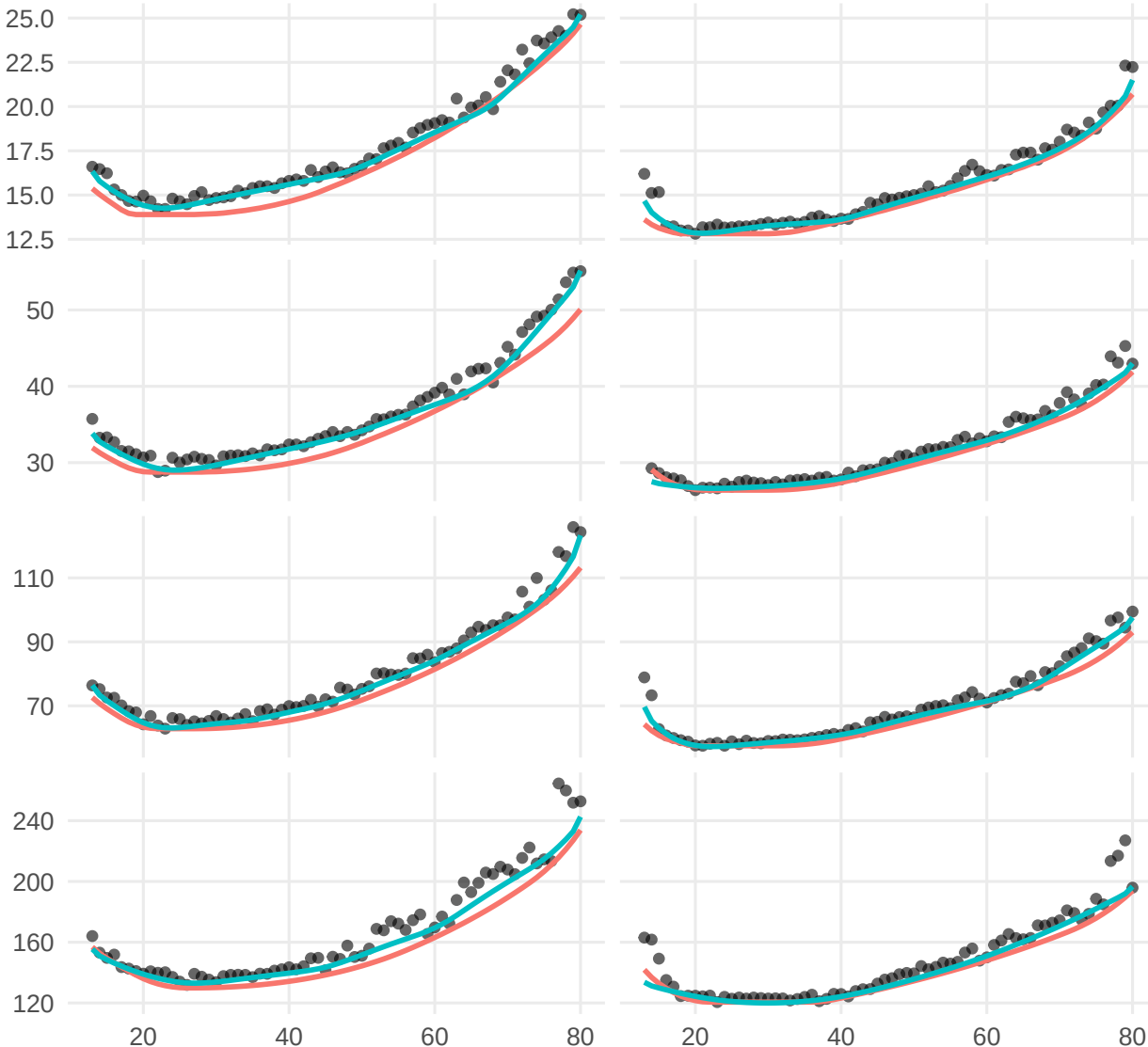
Half Marathon

Marathon

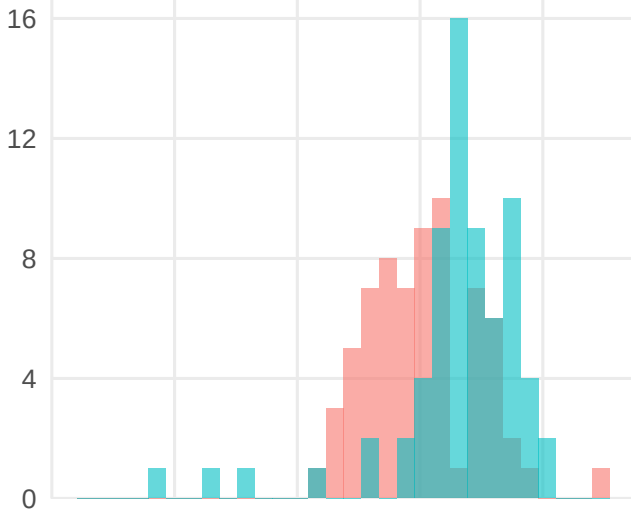
Time (minutes)

Age (years)

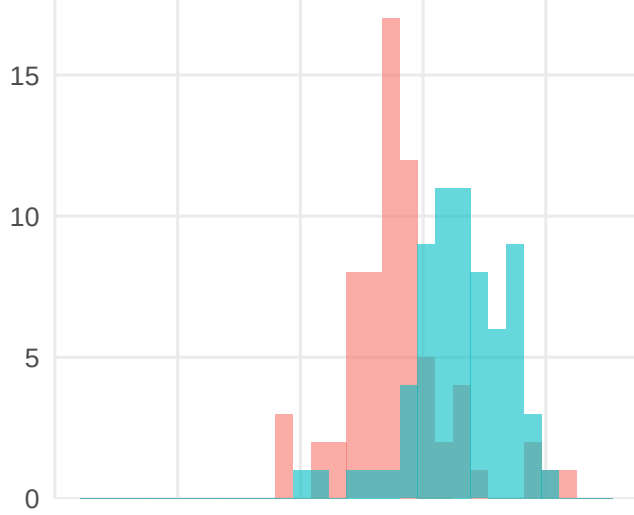
Existing Standard Proposed Standard



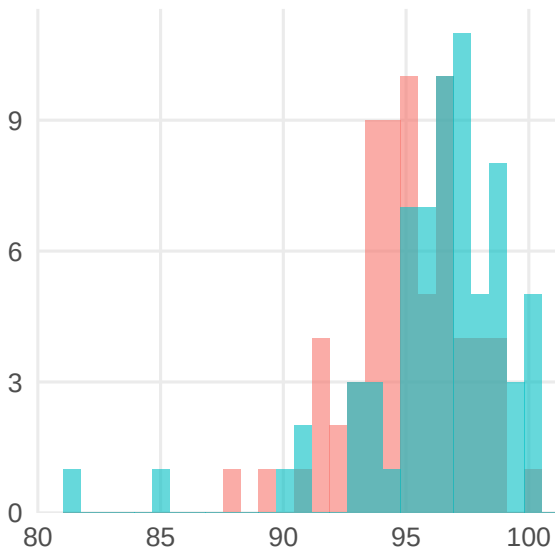
5 km



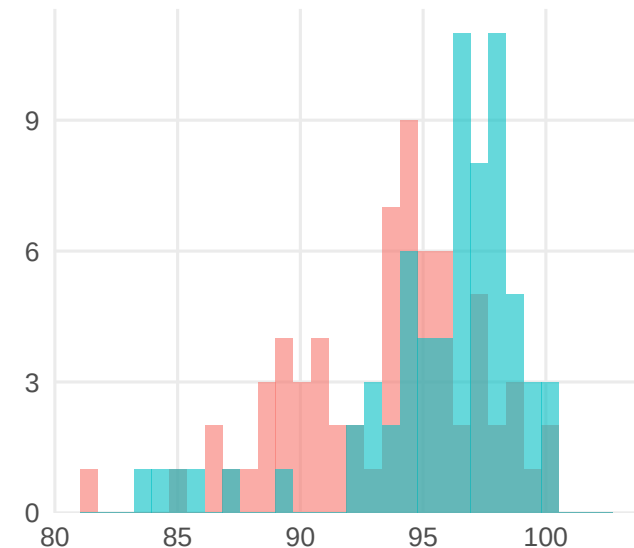
10 km



Half Marathon



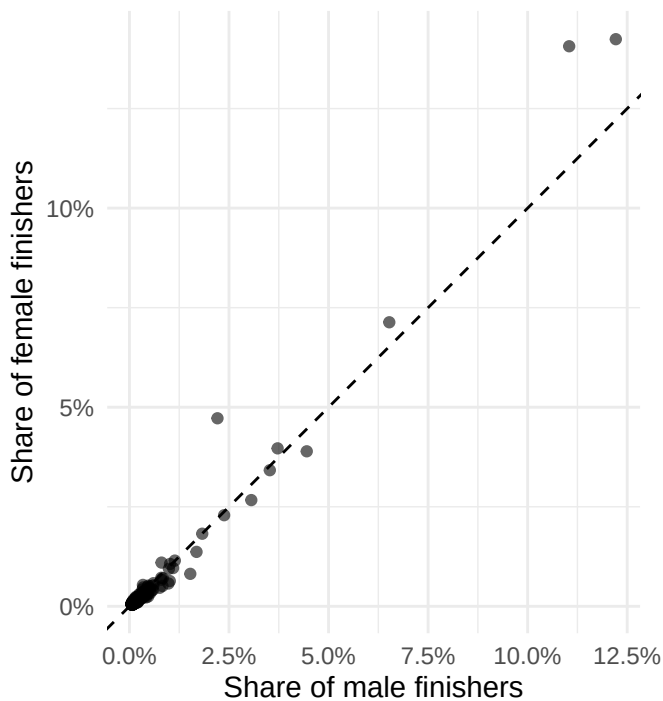
Marathon



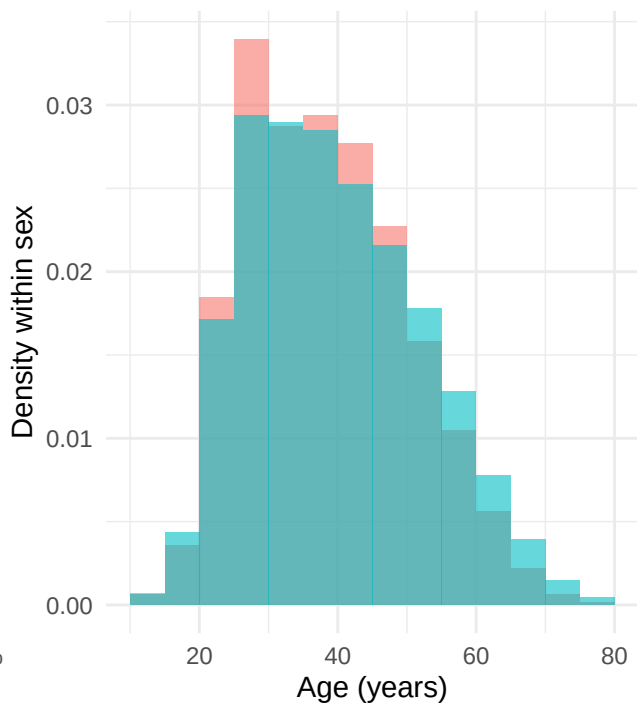
Age grade (%)

Sex Female Male

Participation across races

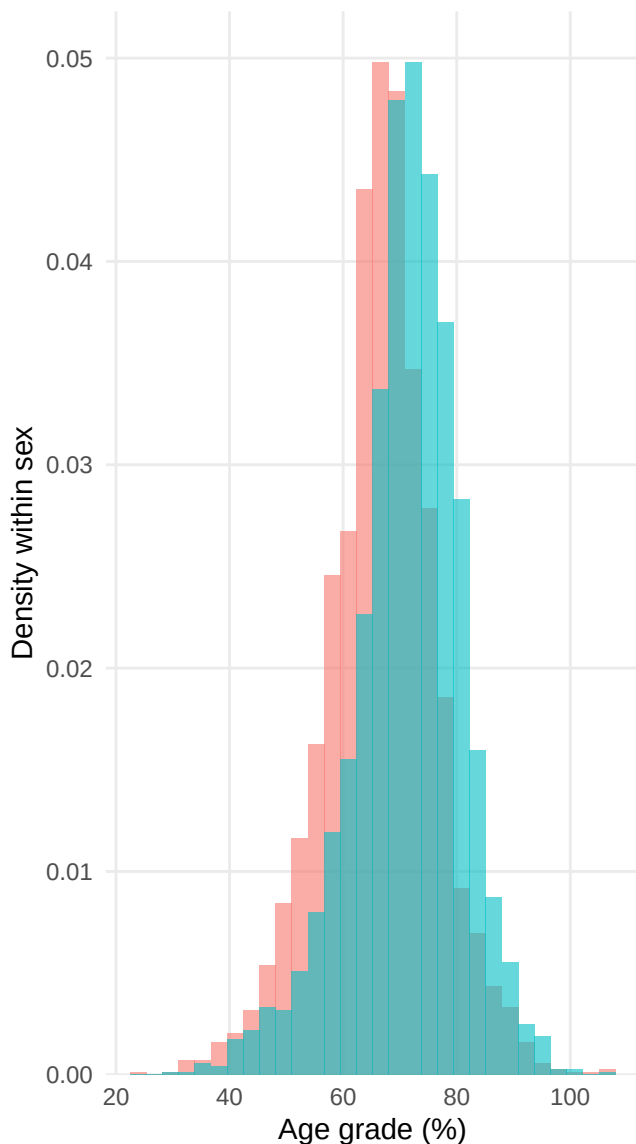


Participation across ages



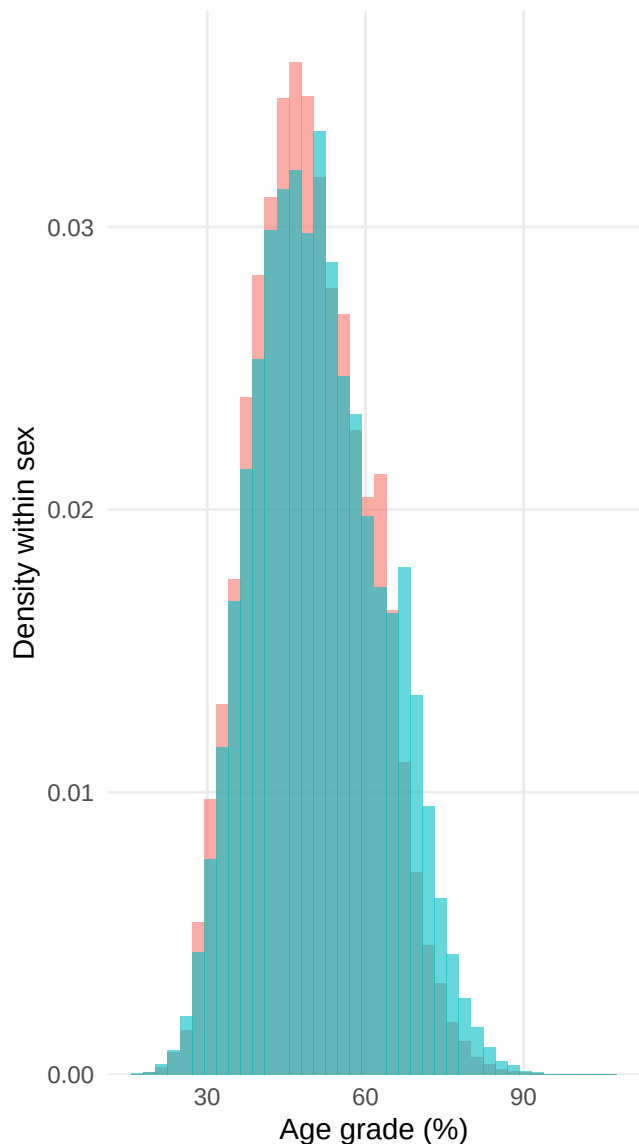
Sex ■ Female ■ Male

# Matched age-group winners



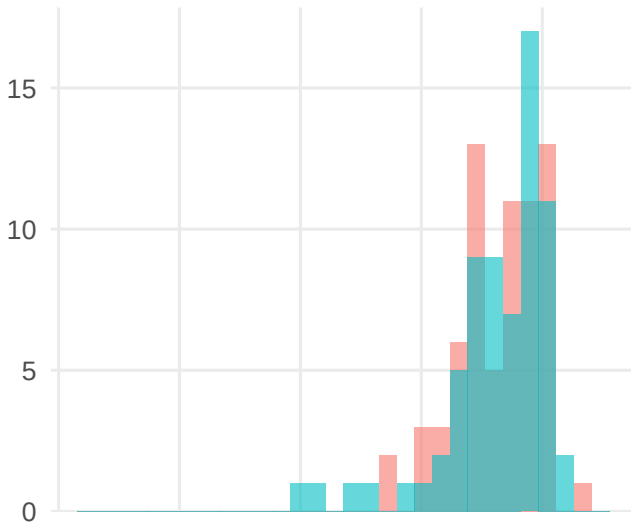
Sex Female Male

# All finishers

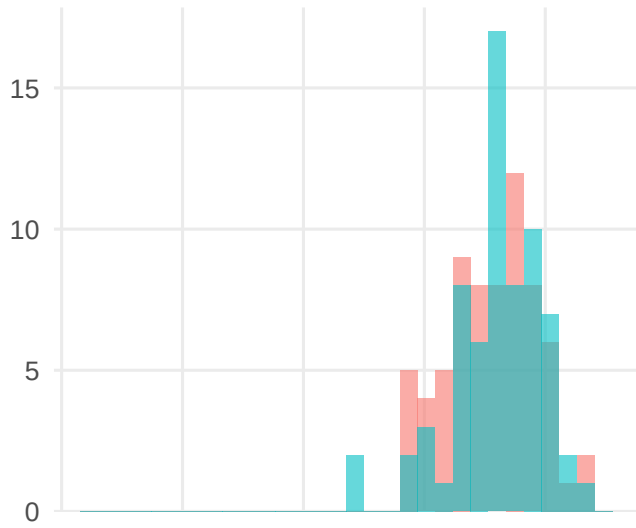


Sex Female Male

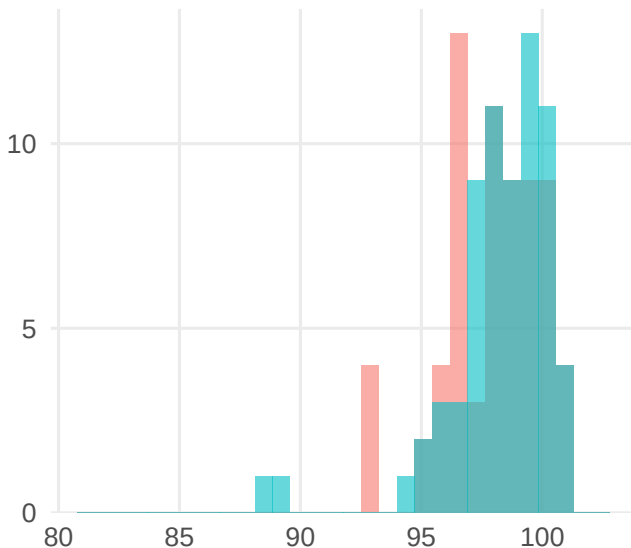
5 km



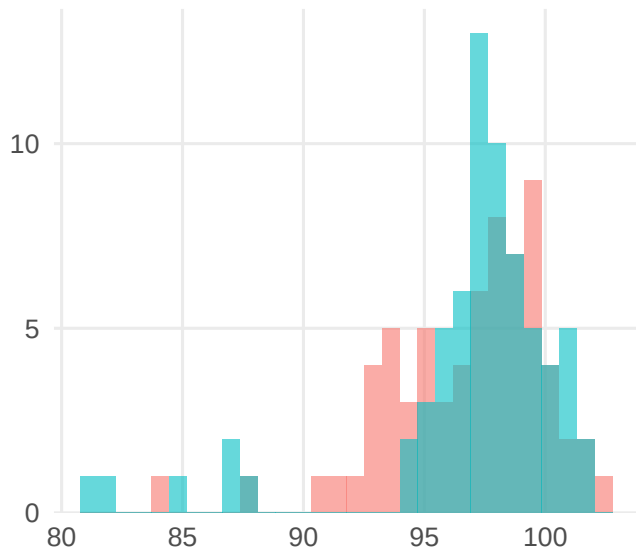
10 km



Half Marathon



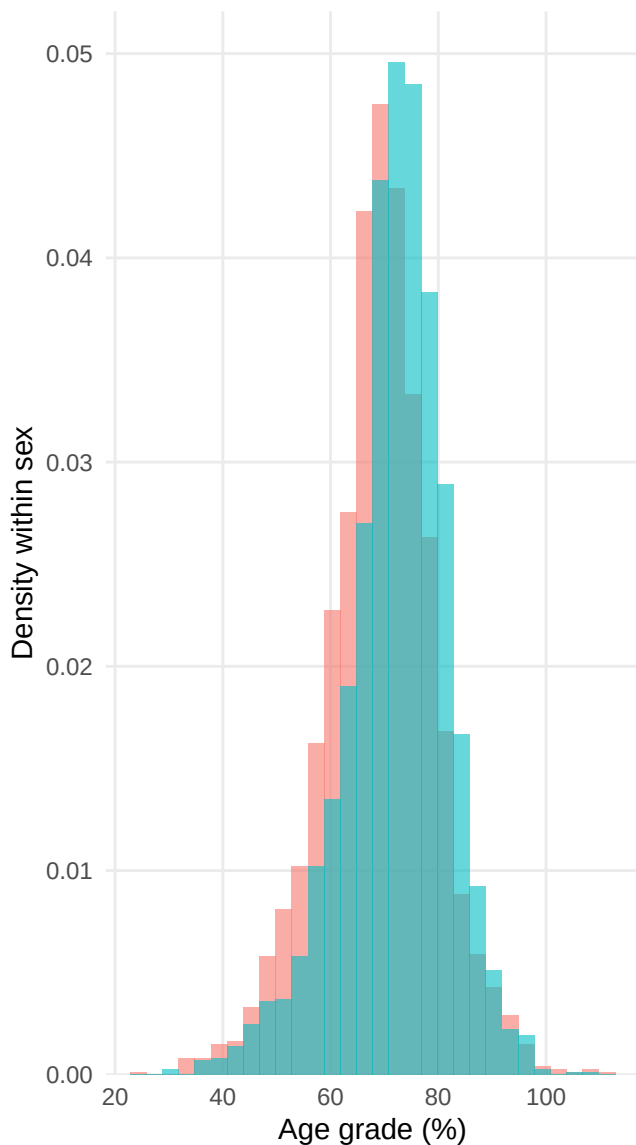
Marathon



Proposed age grade (%)

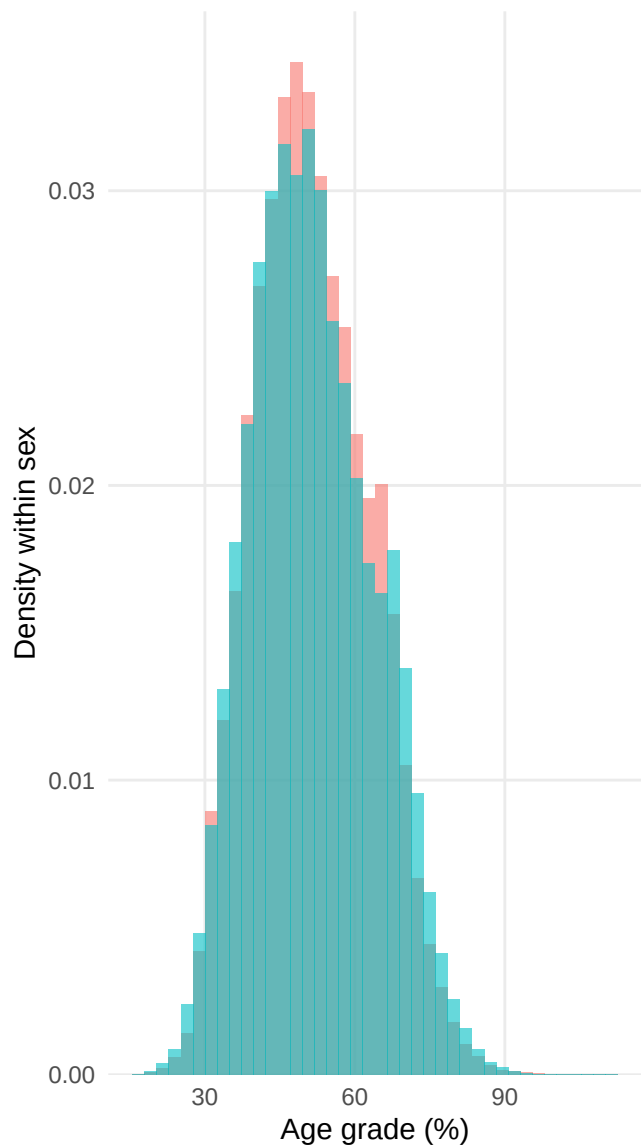
Sex Female Male

# Matched age-group winners



Sex Female Male

# All finishers



Sex Female Male